Marko Tsymbaliuk

tsymma01@gettysburg.edu

markotsymbaluk.com · GitHub · LinkedIn

Summary

Computer Science senior building distributed and real-time systems. Creator of *wikiracing.org* (20,000+ monthly active users; 300,000+ games played) and *CodeCafé* (500+ GitHub stars); Apache Cassandra contributor.

Education

- **Gettysburg College** — Gettysburg, PA
  Bachelor of Science in Computer Science; Minor in Economics; Major GPA: 3.63/4.0 — *Expected Dec 2026*
  - **Honors**: Dean's List (2023, 2024); Presidential Scholar
  - **Coursework**: Computer Networks, Artificial Intelligence, OOP, Data Structures & Algorithms, Databases, Comp Org

Projects

- **wikiracing.org — Multiplayer Wikipedia Racing Platform (20K+ monthly users)** — Live · Demo
  Creator & Sole Maintainer — *Dec 2025 – Present*
  - **Concurrent WebSocket Backend**: Built the Go WebSocket backend behind 300,000+ game starts with per-client read/write goroutines, bounded send queues, heartbeats, and token-bucket rate limits; mutex-guarded rooms coordinate matchmaking, parties, live positions, and reconnects.
  - **Durable, Verified Results**: Built idempotent result ingestion with per-account and guest outboxes, JWT and opaque guest authentication, and a background Go worker that validates each claimed hop against Wikipedia before crediting PostgreSQL player statistics.
  - **Ghost Replay System**: Records cursor and navigation events as timestamped streams; a Go service validates and promotes candidates to Cloudflare R2, then serves them as synchronized ghost opponents.
- **CodeCafé — Collaborative Cloud IDE (500+ ★, Hacker News front page)** — GitHub · Live · Deep Dive
  Creator & Sole Maintainer — *Jan 2025 – Present*
  - **Custom OT Engine (Core)**: Wrote a conflict-free Operational Transformation (OT) engine from scratch (like Google Docs); it reconciles simultaneous edits across users in real time using Java Spring Boot, WebSockets, and Redis Lua scripts.
  - **Cloud IDE Architecture**: Built a browser-based workspace using React, TypeScript, and Monaco Editor that gives users an instant live preview of their HTML/CSS/JS code.
  - **Infrastructure & CI/CD**: Dockerized the app and wired GitHub Actions to run Jest/JUnit tests and deploy the backend to AWS EC2 and the front end to Vercel; Nginx terminates TLS and proxies WebSockets, with Cloudflare for DNS/CDN.
- **Gedis — Redis Server Built from Scratch in Go** — GitHub
  Creator & Lead Developer — *July 2025 – Present*
  - **Storage Engine & Replication**: Built a concurrent 30-command RESP2 server in Go with binary-safe pipelining, typed keys, TTLs, and skip-list-indexed sorted sets; the race-tested implementation adds crash-recoverable AOF persistence, atomic rewrites, and full/partial primary-replica resynchronization via replication IDs, offsets, and a backlog.

Open Source

- **Apache Cassandra** — View Merged Contribution
  Open Source Contributor — *Aug 2025 – Present*
  - **Shipped in Cassandra 5.0.6 & 6.0**: Made complex settings machine-readable as JSON in the settings virtual table so external tools can parse them without custom logic; preserved existing 5.0 behavior behind a system property. Reviewed by two Apache committers.

Experience

- **Software Engineer** — Remote
  Poozle — AI Job-Matching App — *Oct 2025 – Mar 2026*
  - **Vector Recommendation Engine**: Built a Rocchio-based filter on Pinecone that learns from rejections, aggregating embeddings of jobs a user dismissed and suppressing semantically similar listings from later results.
  - **Streaks Progress Tracking**: Shipped a streak system across React Native and Node.js that derives progress lazily from event timestamps on read, rather than updating counters in a nightly batch job.
- **Software Engineering Intern** — Gettysburg, PA
  Gettysburg College — *May 2025 – Aug 2025*
  - **Custom MCP Server & AI Tutor**: Built a Cursor-style coding tutor into DrJava, the editor used in our intro CS courses. Wrote its MCP server from scratch on Java's built-in HTTP server (no frameworks); file-read and directory tools ground the model in the student's real code, streamed over SSE into a Swing panel.
  - **Academic Integrity Platform**: Created a React app for proctored coding exams that tracks keystrokes, monitors window focus, and builds a timeline of student activity.

Programming Skills

Languages	Java · Go · JavaScript · TypeScript · Python · Lua · SQL
Frameworks & Tools	React · Node.js · Spring Boot · FastAPI · React Native · JUnit · Docker · GitHub Actions
Databases & Storage	PostgreSQL · MySQL · Cassandra · Redis · Amazon S3 · ElastiCache
Concepts	Distributed Systems · Event-Driven Architectures · WebSockets · REST · gRPC